Conformal Diagnostics for Robust Computer Vision under Adversarial Conditions

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Abstract

Deep learning detectors deployed in safety-critical systems fail silently under adversarial attack: confidence scores remain high while predictions become unreliable, and no internal signal warns downstream modules that the perceptual input they depend on has been compromised. We present a post-hoc diagnostic framework based on conformal prediction that wraps around any pre-trained object detector—without retraining, without access to internal representations, and without knowledge of the specific attack—to monitor its operational health, quantify degradation severity, and abstain explicitly when output cannot be trusted.

The framework is evaluated on seven detectors spanning four architecture families under five gradient-based attacks, and five natural corruptions on COCO validation set, with a nominal coverage target of $1-\alpha = 0.90$. A selective predictor rejects 82% of adversarial images while retaining 94% of clean ones, with a formal false-abstention bound on clean data. A runtime sliding-window monitor detects adversarial onset within 4–5 images and confirms recovery automatically. A set of conformal diagnostic tools map damage to individual object categories, distinguish localisation degradation from uniform suppression, and provide formal evidence of exchangeability violation. Finally, the coverage gap, a calibrated severity metric grounded in the conformal guarantee, enables direct comparison of adversarial attacks and natural corruptions.

This work present principled diagnostic infrastructure capable of quantifying degradation severity on a formally grounded scale, identifying which perceptual capabilities are affected, distinguishing the nature of the damage, detecting its temporal onset, and confirming recovery, operating entirely at the output level and requiring no assumptions about the image perturbation.

Keywords: conformal prediction, object detection, adversarial robustness, selective abstention, calibration, COCO

1. Introduction

Object detection has become an indispensable component of systems where perception errors carry tangible consequences. Autonomous vehicles rely on detectors to recognise pedestrians, cyclists, and obstacles in real time; medical imaging pipelines use them to localise lesions and anatomical structures; surveillance systems depend on them for threat identification and tracking. Modern architectures—including the YOLO family [10], RT-DETR [11], and DETR [12]—achieve strong performance on standard benchmarks, and their deployment in safety-critical applications continues to accelerate. Yet this growing reliance on learned detectors introduces a vulnerability that remains insufficiently addressed: susceptibility to adversarial perturbations, carefully crafted input modifications that are imperceptible to human observers but cause dramatic detection failures [13, 14, 15].

What distinguishes adversarial failure in object detection from the more extensively studied classification setting is its compound, multi-channel nature. A single adversarial perturbation applied to a detection scene does not merely change a label; it can simultaneously suppress genuine detections (a pedestrian disappears from the scene), fabricate phantom objects (a stop sign appears where none exists), and displace bounding boxes (a vehicle's reported position shifts by several metres). These failures are not accompanied by any drop in confidence scores: the detector continues to report its predictions with high certainty. Downstream systems—path planners, collision avoidance modules,

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diagnostic algorithms, tracking pipelines—receive an output that is internally consistent, confidently stated, and wrong in multiple compounding ways. No internal signal warns them that the perceptual input has been corrupted. This phenomenon, which we term *silent failure*, represents a qualitatively greater hazard than outright system shutdown: a crash triggers fallback mechanisms, but a silently corrupted perception stream leads to decisions that appear well-founded but are based on a distorted view of the world.

The practical implication is clear: safety-critical systems that depend on object detection require a monitoring layer capable of determining, in real time, whether the detector's output can be trusted. The requirements for such a layer are deliberately restrictive, reflecting realistic deployment conditions:

A1. Post-hoc operation. The monitor must wrap around an already-deployed detector without retraining or architectural modification.

A2. Output-only access. The monitor must function by observing only bounding boxes, class labels, and confidence scores—no intermediate features, no gradients, no per-class logit distributions—because the detector is often a vendor-supplied component with a fixed inference API.

A3. Attack-agnostic design. The monitor must make no assumptions about the specific perturbation strategy.

Crucially, such a monitor must provide not merely a binary alarm, but a *formally grounded* assessment: how severely is the system compromised, which perception capabilities remain functional, and when does it recover.

Conformal prediction (CP) [1, 2] is a natural candidate for constructing such a layer, because it satisfies all three constraints by design. It is a distribution-free statistical framework that produces prediction sets with finite-sample coverage guarantees, requiring only that calibration and test data be exchangeable—an assumption weaker than independence that holds under standard evaluation conditions. It operates post-hoc (A1): calibration is a one-time procedure applied to the outputs of any pre-trained model. It requires only output-level observations (A2): nonconformity scores are computed from predictions and ground truth matches, not from internal representations. And it makes no assumption about the perturbation mechanism (A3): the coverage guarantee depends solely on the statistical relationship between calibration and test data.

What makes CP particularly suited to the adversarial monitoring problem—beyond satisfying the access constraints—is the *diagnostic structure* embedded in the guarantee. The guarantee provides not just a binary pass/fail criterion ("does coverage hold?") but a formally referenced *scale*: the gap between the promised coverage and the observed coverage is a calibrated severity metric whose baseline is a mathematical constant ($1-\alpha$), not a tuned threshold. This gap can be decomposed per category, evaluated across spatial overlap thresholds, and tested for statistical significance—producing a rich diagnostic picture from outputs alone.

Recent work has demonstrated CP's utility for detection: Timans et al. [7] applied CP to bounding box calibration, and Angelopoulos and Bates [6] established it as a practical uncertainty quantification tool. However, all existing applications of CP to detection operate exclusively on clean data. The behaviour of conformal prediction under adversarial perturbation—the setting in which uncertainty quantification is most urgently needed—has not been systematically investigated.

We address this gap by developing an adaptive conformal calibrator and a comprehensive diagnostic framework, and evaluating both on seven detectors spanning four architecture families under five gradient-based attacks, APGD with restarts, and five natural corruptions on COCO val2017. During this investigation, we observed a fundamental constraint: conformal coverage after recalibration is bounded by the detector's residual class-correct recall—a ceiling that no output-only method can breach, which we formalise and verify across all twelve model–attack pairs (Cohen's $d > 1.28$, $p < 0.001$). In every case, attempting to recalibrate on adversarial data produces zero improvement, because the attack has already destroyed the detections that recalibration would need to exploit.

Rather than diminishing the case for conformal methods, this observation *sharpens* it. If CP *could* restore coverage, the research agenda would centre on better calibration strategies. The impossibility of recovery redirects attention to what CP *can* provide: a formally grounded infrastructure for measuring how far the system has fallen, identifying which perception capabilities are compromised, characterising the nature of the damage, detecting its temporal onset, and confirming when the system recovers. The framework we present is built on this diagnostic principle.

A further finding emerged from the investigation: adversarial attacks against detectors operate through a fundamentally different mechanism than attacks against classifiers. In classification, every input produces exactly one output, so the adversary must corrupt the class distribution—enlarging prediction sets and creating a detectable distributional signal. In detection, the adversary can suppress outputs entirely, bypassing the distributional channel. This structural distinction explains why CP-based distributional monitoring signals, effective in classification, offer limited value for detection—and motivates the output-level diagnostic design adopted in this work.

The investigation is organised around three research questions:

RQ1. What monitoring and diagnostic capabilities does conformal prediction provide for object detection systems operating under adversarial and naturally corrupted conditions?

RQ2. How do detector architecture, intrinsic calibration quality, and attack strength interact with CP-based monitoring, and what design trade-offs emerge for safety-critical deployment?

RQ3. Can conformal diagnostic tools provide operationally actionable information that goes beyond what confidence-based monitors can offer?

***Contributions*.** We make three contributions that collectively establish conformal prediction as a diagnostic and monitoring infrastructure for adversarially threatened object detection systems.

C1 (Diagnostic and monitoring framework). We propose a post-hoc system that provides three layers of operational awareness. A *selective predictor*, driven by output-level statistics calibrated against the clean operating profile, rejects 82% of adversarial images while retaining 94% of clean ones (6% false-abstention), with a formal false-abstention bound on clean exchangeable data (Proposition 1). A *runtime sliding-window monitor* tracks system health against the conformal coverage target, detecting adversarial onset within 4–5 images under strong attacks and confirming recovery automatically when the attack ceases—without manual reset or domain-specific threshold tuning. The *coverage gap* $\Delta = (1-\alpha) - \mathrm{Cov}_{\mathrm{obs}}$, a calibrated severity metric derived from the conformal guarantee, provides a universal scale for comparing adversarial attacks and natural corruptions, with formal reference points that make severity interpretable without domain knowledge ($\Delta = 0.30$ universally means "30% below the statistical guarantee").

C2 (Conformal diagnostic tools). We demonstrate three analytical capabilities that are unique to the conformal framework and have no analogue in confidence-based monitoring. *Class-conditional coverage* decomposes global performance into per-category coverage values, revealing that small-object classes (handbag, bird, traffic light) lose the conformal guarantee entirely under PGD-20 while person retains 19%—information that enables graduated operational response (targeted fallback for compromised categories rather than total shutdown). *Multi-IoU damage profiling* evaluates coverage across a range of spatial thresholds and distinguishes two qualitatively distinct failure modes: localisation degradation, where surviving detections have degraded spatial accuracy (IoU ratio 0.24 under TOG-V vs 0.45 clean), and uniform suppression, where detections are eliminated proportionally but survivors remain spatially reliable (ratio 0.44 under PGD-20). *Conformal p-values* provide formal statistical evidence of exchangeability violation (KS $p < 0.0001$ for YOLOv8x under PGD-20), constituting a theoretical diagnostic that no confidence monitor can produce.

C3 (Architecture–calibration analysis and structural finding). We uncover a previously unreported 7.2× ECE gap between RT-DETR-L (0.033) and DETR-R101 (0.240) despite both employing transformer decoders, demonstrating that accuracy and calibration are orthogonal properties in modern detectors. We further identify a calibration–monitoring paradox: well-calibrated models degrade gracefully under attack, producing smaller anomaly shifts that are *harder* to detect, creating a design tension between prediction reliability and monitoring sensitivity. Finally, we document the structural distinction between adversarial attack in detection and classification—query suppression versus distribution corruption—which explains the limited effectiveness of distributional monitoring signals and provides principled guidance for monitor design.

Paper structure. Section 2 surveys conformal prediction, adversarial detection methods, and robustness evaluation. Section 3 establishes the theoretical foundations: detection formalism, conformal prediction, coverage notions, adversarial perturbations, and the threat model. Section 4 presents the proposed framework: the adaptive conformal calibrator, the diagnostic and monitoring system, and the conformal analytical tools. Section 5 describes the experimental design, including model and attack selection rationale. Section 6 reports results organised thematically: calibration landscape, adversarial coverage behaviour, selective abstention, runtime monitoring, CP-specific diagnostics, natural corruptions, and signal analysis. Section 7 discusses the implications, the detection–classification structural distinction, and the calibration–monitoring paradox. Section 8 acknowledges limitations, and Section 9 concludes.

2. Related Work

Conformal prediction has attracted growing interest as a framework for uncertainty quantification in visual recognition, owing to its distribution-free coverage guarantee and minimal assumptions. In the classification setting, Romano et al. [5] introduced Adaptive Prediction Sets (APS), which accumulate class labels until a cumulative confidence threshold is met, producing prediction sets that are both valid and adaptive to instance difficulty. Angelopoulos and Bates [6] provided a comprehensive tutorial establishing split conformal prediction as a practical tool for deep learning pipelines. The extension to object detection is more recent and less developed. Timans et al. [7] proposed a two-step procedure that applies separate conformal stages to classification and spatial regression, achieving valid coverage on clean COCO data but producing box margins approximately six times wider than necessary due to the loose coupling between the two stages. Li et al. [8] constructed joint prediction sets over bounding boxes, though without evaluating behaviour under distribution shift. A common limitation of all these methods is the exchangeability assumption: they are designed and evaluated exclusively on clean data, and their behaviour when this assumption is violated—precisely the adversarial setting where uncertainty quantification is most needed—has not been studied.

Several theoretical extensions address distribution shift in conformal prediction. Weighted CP [28] corrects for covariate shift by reweighting calibration scores with the likelihood ratio $P_{\text{test}}/P_{\text{cal}}$, but this ratio is generally unknown under adversarial perturbation. Conformal risk control [29] generalises the coverage guarantee to arbitrary loss functions under comparable reweighting assumptions, controlling the expected value of any monotone loss—including false negative rate and spatial distance—up to $O(1/n)$. Andéol et al. [41] recently extended this to object detection via Sequential Conformal Risk Control (SeqCRC), which applies CRC sequentially to classification and localisation parameters and provides formal guarantees on detection-level risk. SeqCRC represents the closest formal treatment of conformal guarantees for structured detection outputs; however, it operates exclusively on clean data and does not address the adversarial setting or the diagnostic questions (severity quantification, damage mapping, exchangeability testing) that motivate the present work. Our coverage gap can be viewed as an empirical counterpart to CRC's risk control: where CRC guarantees $\mathbb{E}[L] \leq \alpha$ for a chosen loss L under exchangeability, the coverage gap measures $\mathbb{E}[L] - \alpha$ when exchangeability is violated, converting the formal guarantee into a calibrated diagnostic signal. Barber et al. [30] provide the broadest extension, removing exchangeability entirely through online reweighting, yet the worst-case guarantee under arbitrary adversarial shift degenerates to the vacuous interval $[0, 1]$. Gibbs and Candès [33] propose Adaptive Conformal Inference (ACI), which dynamically adjusts the significance level α_t to maintain local coverage under gradual distributional drift. While ACI addresses a related problem, its learning-rate-governed adaptation is designed for smooth, progressive change; the sudden onset characteristic of adversarial attacks—where coverage can collapse within a single image—falls outside its design regime. None of these extensions has been evaluated in the detection setting, and none provides the diagnostic infrastructure (severity quantification, per-class mapping, damage profiling) that our work develops.

A more recent line of work directly targets the recovery of conformal coverage under adversarial perturbation. Gendler et al. [37] introduced Randomly Smoothed Conformal Prediction (RSCP), which integrates randomised smoothing with CP to produce prediction sets with provable coverage under ℓ_2-bounded adversarial perturbation. Yan et al. [38] refined this approach in RSCP+, fixing a flaw in the original RSCP guarantee and improving prediction set efficiency via temperature scaling. Zargarbashi et al. [40] proposed CDF-Aware Sets (CAS), which exploit the CDF structure of scores to achieve tighter robust prediction sets with calibration-time correction. Most recently, Jeary et al. [39] proposed Verifiably Robust Conformal Prediction (VRCP), which leverages neural network verification to recover coverage guarantees under ℓ_1, ℓ_2, and L_∞-bounded perturbations for both classification and regression. These methods represent important progress but differ fundamentally from our setting in two respects. First, all three require internal model access: RSCP/RSCP+ need the score function evaluated under Gaussian noise (requiring multiple stochastic forward passes), while VRCP requires propagating perturbation bounds through the network architecture—incompatible with our output-only constraint (A2). Second, all three are evaluated exclusively on classification tasks (CIFAR-10, CIFAR-100, TinyImageNet) and have not been extended to detection, where the variable-length output and recall ceiling (Section 6.2) create challenges that have no counterpart in classification. Our work is complementary: where RSCP/VRCP seek to *restore* coverage guarantees under attack, we characterise the *diagnostic information* available from the guarantee's violation under output-only constraints.

A substantial body of work addresses the problem of detecting adversarial examples before they cause harm. Feature Squeezing [17] applies input transformations (bit-depth reduction, spatial smoothing) and flags images whose predictions diverge between original and squeezed versions—an effective approach that we adapt as an output-level baseline. ODIN [20] improves out-of-distribution detection by combining temperature scaling with small input perturbations, but requires gradient computation through the model. Mahalanobis-based detection [21] fits class-conditional Gaussians in intermediate feature spaces and uses Mahalanobis distance as a detection score, achieving strong performance but requiring access to all hidden layers. Local Intrinsic Dimensionality (LID) [18] exploits the observation that adversarial examples occupy lower-dimensional subspaces of the feature manifold, again requiring internal representations. Deep k-NN [22] measures embedding distances, and MagNet [19] relies on autoencoder reconstruction error. Spectral methods [26] analyse covariance structure in hidden activations. All of these approaches achieve their detection capability through access to the model's internal representations—a requirement incompatible with the output-only constraint (A2) that defines our deployment setting.

Test-time augmentation (TTA) [34] offers an alternative that is model-agnostic: it evaluates prediction consistency across multiple augmented views of the input. This approach respects the output-only constraint but incurs a latency penalty of 5–10×, making it impractical for real-time detection pipelines operating at 30 FPS. A lightweight variant using 2–3 views would reduce the overhead to 2–3×, potentially offering a pragmatic middle ground for applications where moderate latency is acceptable. We do not include TTA as an experimental baseline due to the real-time constraint central to our deployment setting, but acknowledge it as the most natural comparison point for relaxing the single-pass assumption and discuss it further in Section 8. In streaming deployment scenarios, frame-to-frame temporal consistency—measuring detection stability across consecutive frames via metrics such as track fragmentation or IoU overlap between successive outputs—provides another output-level anomaly signal. Such metrics exploit the temporal smoothness prior that adversarial perturbations applied independently per frame are likely to violate. Our sliding-window monitor (Section 4.2) captures a related signal through windowed coverage tracking, but a dedicated temporal consistency score could complement the per-image anomaly score; we leave this integration to future work. More recently, Lekeufack et al. [35] integrated conformal coverage with rejection decisions optimised through ROC analysis, demonstrating the potential of CP-based abstention, though their work addresses classification rather than detection. Rossolini et al. [36] provide a comprehensive survey of runtime monitors for perception systems and highlight a critical gap: formal guarantees on false alarm rates are rare among methods operating solely on model outputs. Our work addresses this gap directly, providing a formal false-abstention bound within a single-pass, output-only framework.

The landscape of gradient-based attacks on object detectors spans a range of strategies and computational budgets. FGSM [13] computes a single gradient step, providing a cheap but often weak baseline. PGD [14] iterates multiple projected gradient steps and has become the standard adversarial evaluation tool. DAG [15] jointly optimises classification and localisation losses, and TOG [16] directly targets the objectness mechanism. Together, these attacks reduce detection mAP by 55–90% under standard L_∞ budgets. A critical lesson from recent benchmarking efforts, particularly AutoAttack [25], is that perceived robustness is frequently an artefact of weak or improperly configured attacks—a phenomenon we encounter directly with DETR, where the gap between PGD-20 and APGD results is a factor of twenty. For transformer-based detectors specifically, Luo et al. [31] and Chen et al. [32] have identified the decoder cross-attention mechanism as a primary vulnerability, explaining why classification-only losses without restarts fail to produce meaningful degradation. Bhojanapalli et al. [27] have shown that vision transformers exhibit qualitatively different robustness properties from CNNs, further motivating our multi-family evaluation.

On the defence side, adversarial training [14] remains the dominant empirical approach, incorporating adversarial examples into the training loop to build robustness. It provides measurable protection but requires full retraining at substantial computational cost and typically degrades clean-data performance. Certified defences, notably randomised smoothing [24], offer provable robustness guarantees but at the expense of both accuracy and inference speed. Neither approach is applicable post-hoc to an already-deployed detector. Conformal prediction occupies a complementary niche: it requires no retraining, no architectural modification, and no knowledge of the attack, operating entirely as a post-hoc wrapper. It does not improve the detector's robustness in the adversarial-training sense—it cannot recover detections that the model fails to produce—but it provides a diagnostic and monitoring layer that adversarial training and certified defences do not address.

3. Background and Theoretical Foundations

This section introduces the formal apparatus that underlies the proposed framework. We begin with the detection setting and the notion of calibration trust (Section 3.1), then develop the conformal prediction machinery and its adaptation to detection (Section 3.2), define the coverage quantities that serve as evaluation criteria (Section 3.3), and close with the adversarial perturbation landscape and the threat model that delimits the defender's design space (Section 3.4).

3.1. Object Detection and Calibration

An object detector $f\colon \mathcal{X} \rightarrow 2^{\mathcal{D}}$ maps an image x to a variable-size set of detections $\{(b_i, c_i, s_i)\}_{i=1}^{N_x}$, each comprising a bounding box $b_i \in \mathbb{R}^4$, a class label $c_i \in \{1, \ldots, K\}$, and a confidence score $s_i \in [0, 1]$. Two complementary metrics characterise performance. Mean Average Precision, mAP@[0.5:0.95], averages precision across ten IoU thresholds and all categories, providing the primary benchmark measure of detection quality. Average Recall, AR@100, counts the fraction of ground truth objects matched by the top-100 proposals *before* any confidence filtering. Because AR@100 reflects the detector's raw capacity to generate proposals that spatially overlap with ground truth, it represents a natural ceiling on what any post-hoc method can achieve: proposals that were never generated cannot be recovered by downstream processing.

Whether a detector's confidence scores can be *trusted* is a question distinct from whether its detections are accurate. A perfectly calibrated detector assigns confidence s to detections of which exactly a fraction s are true positives; in practice, detectors are systematically over- or under-confident. The degree of miscalibration is captured by the Expected Calibration Error, which partitions detections into B equal-width confidence bins and averages the gap between empirical accuracy and stated confidence:

$$\mathrm{ECE} = \sum_{m=1}^{B} \frac{|I_m|}{N} \left|\mathrm{acc}(m) - \mathrm{conf}(m)\right| \tag{1}$$

Calibration quality has a direct but often overlooked consequence for uncertainty quantification: a well-calibrated detector's scores already carry reliable information about prediction quality, reducing the amount of statistical correction any downstream framework must apply. This relationship becomes particularly relevant when conformal margins are computed, as described next.

3.2. Conformal Prediction for Object Detection

Conformal prediction [1, 3] is a distribution-free framework that constructs prediction sets with finite-sample coverage guarantees. Its central requirement is *exchangeability*: the joint distribution of calibration and test observations must be invariant to permutation. This assumption is weaker than independence and holds whenever both sets are drawn from a common distribution. It is satisfied under standard evaluation conditions and violated under adversarial perturbation—a distinction that is central to the present work.

Definition 1 (Split Conformal Prediction)**.** *Given calibration nonconformity scores $\{V_i\}_{i=1}^n$ and miscoverage level $\alpha \in (0, 1)$, define the conformal quantile*

$$\hat{q} = \mathit{Quantile}_{\lceil(1-\alpha)(n+1)\rceil/n}(\{V_i\}_{i=1}^n) \tag{2}$$

and the prediction set $C_\alpha(x)=\{y : V(x, y) \leq \hat{q}\}$. Under exchangeability:

$$1-\alpha \;\leq\; \mathbb{P}[Y_{n+1} \in C_\alpha(X_{n+1})] \;\leq\; 1-\alpha + \tfrac{1}{n+1} \tag{3}$$

The lower bound is exact and distribution-free; the upper bound reflects finite-sample discretisation.

Applying this guarantee to detection requires specifying what constitutes an observation and how nonconformity is measured. We adopt a per-detection formulation. Given a detection (b_i, c_i, s_i) matched to a ground truth annotation (b_j^*, c_j^*) via IoU≥0.5 with $c_i{=}c_j^*$, we define two scores. The classification nonconformity $V_{\mathrm{cls}}{=}1{-}s_i$ measures how far confidence falls below certainty; its quantile yields a threshold $\tau{=}1{-}\hat{q}_{\mathrm{cls}}$ below which detections are filtered. The localisation nonconformity

$$V_{\mathrm{loc}} = \|b_i - b_j^*\|_1 \,/\, \mathrm{area}(b_j^*) \tag{4}$$

measures size-normalised box residuals, ensuring that a 5-pixel error is penalised more heavily on a small object than on a large one. Its quantile $\hat{q}_{\mathrm{loc}}$ defines a spatial margin: the conformal set for a detected box. Following Conformalized Quantile Regression [4], the localisation quantile can be further stratified by confidence bin, producing tighter margins for high-confidence detections where localisation is typically more precise.

A critical difference from conformal prediction in classification is that detectors produce a *variable number* of outputs per image. This variability has profound implications under adversarial conditions: an attack that suppresses the proposal mechanism can reduce N_x from dozens to near zero, eliminating the detections on which conformal sets would be constructed. This channel of degradation—output suppression rather than output corruption—is distinct from the classification setting, where every input always produces exactly one prediction.

Remark 1 (Observation unit and coverage interpretation)**.** *The CP guarantee (Definition 1) applies per matched (detection, ground truth) pair: a randomly drawn pair falls within the conformal set with probability at least* $1-\alpha$*. The "post-threshold coverage" used throughout this paper—the fraction of ground truth objects matched by at least one surviving conformal detection—is a related but distinct, image-level quantity that inherits the per-detection guarantee only approximately, because detection counts vary per image and false positives contribute calibration scores without appearing in the coverage count. We adopt this metric as the operationally relevant measure (whether each real object is detected) and note that the coverage gap's diagnostic value does not require the per-detection guarantee to transfer exactly: it requires only that* $\Delta \approx 0$ *on clean data and* $\Delta > 0$ *under attack, which we verify empirically.*

3.3. Coverage Quantification

Two notions of coverage play distinct roles in the analysis. *Pre-threshold coverage* counts the fraction of ground truth objects matched by any class-correct, spatially overlapping detection from the full output set, before conformal filtering is applied. It is equivalent to the class-correct recall $\mathrm{AR}(f, \delta)$ and represents a hard ceiling: no post-hoc operation—whether re-thresholding, re-weighting, or spatial expansion—can create coverage for objects that the detector failed to produce proposals for. *Post-threshold coverage* counts matches only among detections that survive the conformal confidence threshold τ; it is the operational metric targeted by the guarantee in Eq. (3) and satisfies $\mathrm{Cov}_{\mathrm{post}} \leq \mathrm{Cov}_{\mathrm{pre}}$ by construction.

The gap between these two quantities motivates the central diagnostic metric of this work.

Definition 2 (Coverage Gap)**.** *The coverage gap measures the deviation of observed conformal coverage from the nominal guarantee:*

$$\Delta_\alpha(\delta) = (1-\alpha) - Cov_{post}(f_{CP}, \delta) \tag{5}$$

On clean exchangeable data, $\Delta \leq 0$ *by Eq.* (3)*. Under perturbation,* $\Delta > 0$ *quantifies the degree of guarantee violation on a universal scale:* Δ=0.30 *means "coverage is 30 percentage points below the statistical guarantee," interpretable without domain-specific knowledge.*

The gap admits a natural severity interpretation, which we operationalise as four strata. Values below 0.10 fall within the finite-sample tolerance of the conformal guarantee (the $O(1/n)$ term in Eq. 3). Values between 0.10 and 0.30 indicate meaningful degradation but partial functionality. Values between 0.30 and 0.60 indicate severe compromise, and values above 0.60 indicate near-total loss of the coverage guarantee. These thresholds are operational conventions informed by the guarantee's structure—the first band corresponds to the finite-sample discretisation range, the last to near-complete guarantee violation—rather than formal derivations.

They serve as interpretive reference points that are consistent across attacks, models, and datasets, but alternative stratifications could be chosen for specific deployment contexts. As a sensitivity check: shifting all band boundaries by ±0.05 (e.g., using 0.15/0.35/0.65 instead of 0.10/0.30/0.60) reclassifies at most 2 of the 12 model–attack pairs in Table 2 into adjacent bands, confirming that the diagnostic conclusions are not sensitive to the precise boundary placement.

3.4. Adversarial Perturbations in Object Detection

An adversarial perturbation is a modification δ added to an input image x such that a model's output on $x+\delta$ differs substantially from its output on x, while δ remains imperceptible to a human observer. Imperceptibility is formalised through a norm constraint $\|\delta\|_p \leq \varepsilon$; the L_∞ norm, bounding the maximum change to any single pixel, has become the standard choice in robustness evaluation, with ε=8/255 (≈3% of the full dynamic range) as the established benchmark budget [14, 25].

Object detection presents a fundamentally richer attack surface than classification. Because a detector produces a variable-length set of predictions, a single perturbation can affect the output through three distinct channels: *suppression* of true detections, *fabrication* of false detections, and spatial *displacement* of bounding boxes. These channels may activate simultaneously, and the detector's confidence scores can remain high throughout, providing no intrinsic warning that the output has been compromised. This compound failure mode—internally consistent, confidently reported, and wrong in multiple ways—is what distinguishes adversarial failure in detection from the single-label corruption that characterises classification attacks.

Perturbations are constructed via gradient-based optimisation. The attacker maximises a loss $\mathcal{L}(f(x+\delta), y)$ subject to $\|\delta\|_\infty \leq \varepsilon$; the choice of $\mathcal{L}$ determines which damage channel is targeted. Classification losses corrupt predicted labels, localisation losses shift bounding box coordinates, and objectness losses suppress the detection mechanism itself. Attacks further vary in iteration count—from single-step methods [13] to multi-step optimisation over tens or hundreds of iterations [14, 15, 16]—and in the use of random restarts to escape weak local optima. The interaction between attack configuration and apparent robustness is well documented: the absence of restarts can yield dramatically misleading robustness estimates, particularly for architectures with complex, non-convex loss landscapes [25].

4. Methodology

The theoretical foundations of Section 3 provide three building blocks: a conformal guarantee that holds under exchangeability (Definition 1), nonconformity scores that adapt this guarantee to detection (Eqs. 4–2), and a coverage gap that quantifies guarantee violation on a universal scale (Definition 2). The adversarial perturbation model (Section 3.4) establishes that attacks can degrade detection through suppression, fabrication, and displacement simultaneously, while the output-only constraint (A2) restricts the defender to observing bounding boxes, class labels, and confidence scores.

We now describe how these elements are assembled into an operational framework. The method comprises three components: an adaptive conformal calibrator that establishes the statistical baseline on clean data (Section 4.1), a diagnostic and monitoring system that detects and quantifies degradation in real time (Section 4.2), and a set of conformal analytical tools that characterise the nature of the damage (Section 4.3). All components are calibrated exclusively on clean data, require no model modification, and respect the output-only constraint.

4.1. Adaptive Conformal Calibration

The calibration process translates the abstract conformal guarantee of Definition 1 into a concrete set of operating parameters for the detector. Given a calibration set of images with ground truth annotations, we first run the detector and match each prediction to ground truth using IoU≥0.5 with class-correct matching, following a greedy assignment that prioritises higher-IoU pairs. Three categories of detections arise from this matching: *true positives* (matched to a ground truth object), *false positives* (no matching ground truth), and *missed objects* (ground truth with no matching detection). Only true-positive pairs contribute nonconformity scores to the calibration set; false positives are excluded because they have no ground truth against which to measure nonconformity, and missed objects are excluded because no detection exists to evaluate. This design choice means that the calibration distribution reflects the detector's behaviour on objects it *does* detect, and the conformal threshold τ is tuned to control quality among detected objects.

Objects that the detector fails to propose entirely are outside the scope of the per-detection guarantee—a limitation formalised as the recall ceiling in Proposition 2. A further consequence of calibrating on true positives only is that the resulting thresholds reflect the detector's *clean* false-positive behaviour. Under adversarial attack, the false-positive distribution can shift—for instance, DETR's fixed-query mechanism generates spurious high-confidence detections that would not appear in the clean calibration set (Section 6.5). The conformal threshold τ does not account for this shift, which is why the coverage gap under attack is interpreted as a diagnostic signal rather than a formal guarantee violation.

From these matched pairs, we compute the two nonconformity scores introduced in Section 3.2: the classification score $V_{\text{cls}} = 1 - s_i$, which measures distance from certainty, and the localisation score V_{loc} (Eq. 4), which measures size-normalised box residuals.

Five conformal quantities are derived from these scores. The APS confidence threshold τ [5] is the quantile of classification nonconformity scores at level $1-\alpha$; detections with $s_i < \tau$ are filtered as insufficiently reliable. The global box margin $\hat{q}$ is the corresponding quantile of localisation scores, defining the spatial extent of the conformal prediction set around each detected box. To account for the well-known dependence of localisation precision on object size, we partition detections into three area categories following the COCO definitions (small: $< 32^2$ px, medium: 32^2–96^2 px, large: $> 96^2$ px) and compute separate size-normalised margins per bin. Following CQR [4], we further stratify the localisation quantile by confidence bin, producing tighter margins for high-confidence detections—where the detector's spatial predictions are empirically more precise—and wider margins for uncertain ones. Finally, class prediction sets accumulate plausible labels beyond the top-1 class for each detection, extending the conformal guarantee to the classification dimension. The calibration procedure is summarised in Algorithm 1.

Its output—a tuple $(\tau, \hat{q}, \text{bin quantiles})$—defines the operating envelope within which the conformal coverage guarantee of Eq. (3) holds.

Throughout this paper, we set $\alpha = 0.10$, yielding a nominal coverage target of $1-\alpha = 0.90$. All coverage values, gap computations, severity bands, and selective predictor thresholds are defined relative to this target.

Departures from this envelope, measured by the coverage gap, form the basis of the diagnostic system described next.

Algorithm 1 Adaptive Conformal Calibration

Require: Predictions $\{(b_i, c_i, s_i)\}$, ground truth $\mathcal{G}$, level α=0.10

1: Match predictions to GT (IoU≥0.5, class-correct)
2: Compute V_i^{cls} and V_i^{loc} for matched pairs
3: $\tau \leftarrow 1 - \text{Quantile}_{1-\alpha}(\{V_i^{\text{cls}}\})$
4: $\hat{q} \leftarrow \text{Quantile}_{1-\alpha}(\{V_i^{\text{loc}}\})$
5: Stratify $\hat{q}$ by size bin and confidence bin
6: **return** $(\tau, \hat{q}, \text{bin\_quantiles})$

4.2. Diagnostic and Monitoring System

The diagnostic and monitoring system is the central methodological contribution of this work. Its design reflects a principle that emerged from the interplay between the conformal guarantee and the adversarial threat model: since the coverage guarantee of Eq. (3) presupposes exchangeability, and adversarial perturbation violates exchangeability, the guarantee will be broken under attack. The question is not whether it breaks, but *how informatively* the breakage can be characterised. The system addresses this question through three components, each targeting a distinct operational need: severity quantification, image-level abstention, and temporal monitoring.

The coverage gap $\Delta = (1-\alpha) - \text{Cov}_{\text{post}}$ (Definition 2) converts the binary question "does the guarantee hold?" into a continuous signal "by how much is it violated?" Three properties make this metric operationally superior to standard measures such as mAP drop. Its baseline $1-\alpha = 0.90$ is not a threshold chosen by the operator but the formal conformal guarantee, so $\Delta = 0.30$ is universally interpretable as "30% below the statistical guarantee" without domain context. The same scale applies to adversarial attacks, natural corruptions, and distribution shifts, enabling direct cross-comparison of fundamentally different degradation sources. And the target $\Delta = 0$ is a mathematical consequence of Eq. (3), making positive deviation a formal violation rather than a subjective judgment.

The severity bands defined in Section 3.3 ($\Delta < 0.10$: within guarantee; 0.10–0.30: degraded; 0.30–0.60: severe; ≥ 0.60: destroyed) are anchored to the guarantee structure and do not require tuning against specific attacks or datasets.

Selective prediction with formal abstention. While the coverage gap quantifies degradation at the population level, the selective predictor operates at the individual image level, implementing the principle that an explicit "I don't know" is safer than a confident wrong answer. For each image x, we compute a composite anomaly score from three output-level statistics:

$$A(x) = w_1\, z_{\text{hc}}(x) + w_2\, z_{\max}(x) + w_3\, z_{\mu}(x) \tag{6}$$

Here z_{hc} is the z-score of the high-confidence detection count ($s > 0.1$) relative to the clean calibration baseline, $z_{\max}$ is the z-score of maximum confidence, and z_μ the z-score of mean confidence. All z-scores are computed against the mean and standard deviation observed on the calibration set. The weights ($w_1 = 0.4$, $w_2 = 0.3$, $w_3 = 0.3$) are fixed a priori and reflect the empirical observation—confirmed by ablation in Section 6.7—that detection count is the single most informative statistic under strong attacks, because the suppression channel (Section 3.4) reduces the number of detections before it degrades their confidence.

The anomaly threshold τ_A is set as the $(1-\alpha)$-quantile of $\{A(x_i)\}_{i=1}^n$ on the clean calibration set. The predictor outputs $f_{\text{CP}}(x)$ when $A(x) \leq \tau_A$ and abstains otherwise. This construction yields a formal guarantee on clean data.

Proposition 1 (False-Abstention Bound)**.** *If* $\tau_A = \textit{Quantile}_{1-\alpha}(\{A(x_i)\}_{i=1}^n)$ *on clean calibration data, then*

$$P[A(X_{test}) > \tau_A \mid clean] \leq \alpha \tag{7}$$

Three properties distinguish this from ad-hoc confidence thresholding, though the construction remains a calibrated heuristic rather than a deep conformal object. The false-abstention rate is formally bounded rather than empirically estimated—but this guarantee follows straightforwardly from quantile thresholding on the exchangeable calibration distribution and does not require conformal machinery beyond the exchangeability assumption. The threshold is derived from the calibration distribution rather than tuned against adversarial examples, which are unavailable in deployment. And the score components are causally linked to the channels through which adversarial attacks degrade detection: strong attacks collapse the detection count (captured by z_{hc}), weak attacks depress confidence (captured by $z_{\max}$ and z_μ), and both pathways feed the composite. The fixed weights (w_1=0.4, w_2=0.3, w_3=0.3) are set a priori and validated by ablation (Section 6.7);

a learned weighting could improve performance but would introduce a tuning step that we deliberately avoid to preserve the fully calibration-driven design. We note that the fixed-weight composite has not been tested against adaptive adversaries that explicitly target the anomaly score $A(x)$—for instance, an adversary that suppresses detections while maintaining the count statistic near the clean baseline. Such an adversary would need to simultaneously fool the detector (producing wrong outputs) and preserve the output-level statistics on which the monitor relies, a dual objective that imposes a non-trivial constraint on the attack. Evaluating robustness to adaptive adversaries is an important direction noted in Section 8, and the modular design of the anomaly score (Eq. 6) allows additional components—such as frame-to-frame consistency or prediction volatility—to be incorporated without altering the calibration procedure.

Remark 2 (Scope of the formal guarantee)**.** *The false-abstention bound holds on exchangeable (clean) data only. Under adversarial shift, exchangeability is violated and no formal bound on detection or abstention rates applies—the coverage guarantee, the false-abstention rate, and all derived quantities become empirical observations without formal backing. The selective predictor is therefore an empirical diagnostic tool with formal grounding on clean data, not a certified safety barrier.*

Algorithm 2 Selective Prediction at Inference

Require: Image x, calibration $(\tau, \hat{q}, \text{bins})$, threshold τ_A

```
1:  Compute detections D = f(x)
2:  Compute A(x) via Eq. (6)
3:  if A(x) > τ_A then
4:      return ABSTAIN
5:  end if
6:  for (b_i, c_i, s_i) ∈ D do
7:      if s_i ≥ τ then
8:          Emit detection with conformal set (b_i ± q̂_bin, C_i)
9:      end if
10: end for
```

The full inference procedure is detailed in Algorithm 2. At test time, the framework first evaluates the anomaly score to decide whether to abstain at the image level, then applies conformal filtering and margin construction to the surviving detections.

Runtime sliding-window monitor. The selective predictor and coverage gap operate, respectively, at the image and population levels. The runtime monitor addresses the temporal dimension: in a streaming deployment, the relevant questions are *when* does the system become compromised and *when* does it recover.

A sliding window of W consecutive images tracks the empirical coverage within the window. When ground truth is available (offline evaluation or labelled validation streams), coverage is measured directly. In deployment, the detection count serves as a label-free proxy; the strong correlation between count and coverage collapse (DeLong AUROC = 0.926 for YOLOv8x under PGD-20) justifies this approximation. An alert fires when the windowed metric drops more than 15% below the $1-\alpha = 0.90$ target—a threshold calibrated at 1.5× the maximum per-window deviation observed on clean data, ensuring zero false alarms for $W \geq 15$.

The $1-\alpha$ target serves as a principled alert baseline. Process control methods such as CUSUM and Shewhart charts require manual specification of control limits; the conformal framework derives its reference from the coverage guarantee, eliminating domain-specific tuning. Recovery is confirmed when the windowed metric returns within 10% of the pre-alert baseline for one full window width, preventing premature de-alerting from transient fluctuations and correctly distinguishing temporary attacks from persistent system degradation.

4.3. Conformal Diagnostic Measures

The conformal guarantee provides not only a binary pass/fail criterion but also a rich internal structure that can be decomposed along several dimensions to characterise degradation. We define three diagnostic measures that exploit this structure and have no counterpart in confidence-based monitoring systems.

Class-conditional coverage decomposition. The post-threshold coverage of Definition 2 aggregates across all K object categories, which can conceal severe heterogeneity in how an attack affects different classes. We decompose this aggregate into per-class coverage values. Let $\mathcal{G}_k = \{j : c_j^* = k\}$ denote the set of ground truth objects of class k, and let $\mathcal{M}_k(\delta) \subseteq \mathcal{G}_k$ denote those matched by at least one surviving conformal prediction under perturbation δ. The class-conditional coverage is

$$\mathrm{Cov}_k(\delta) = |\mathcal{M}_k(\delta)| \,/\, |\mathcal{G}_k| \tag{8}$$

and the class-conditional gap is $\Delta_k = (1-\alpha) - \mathrm{Cov}_k$. Categories with $\Delta_k \geq 0.60$ have lost the conformal guarantee entirely; those with $\Delta_k < 0.10$ retain it. This decomposition enables graduated operational response: rather than treating all perception as uniformly compromised, a system can identify which categories remain functional and selectively disengage only those functions that depend on categories whose coverage has collapsed.

Multi-IoU coverage profiling. Standard evaluation assesses coverage at a single spatial overlap threshold (IoU = 0.50). However, the *shape* of coverage across multiple thresholds encodes information about the nature of the damage that a single threshold cannot capture. We evaluate coverage at four thresholds $\mathcal{T} = \{0.25, 0.50, 0.75, 0.90\}$ and define the IoU ratio

$$r(\delta) = \mathrm{Cov}_{@0.90}(\delta) \,/\, \mathrm{Cov}_{@0.25}(\delta) \tag{9}$$

as a summary statistic for the profile shape. The ratio distinguishes two qualitatively distinct failure modes. When $r(\delta) \approx r(0)$ (i.e., the adversarial ratio is close to the clean ratio), detections are eliminated proportionally across all IoU levels: this is *uniform suppression*, in which the number of detections decreases but the spatial accuracy of survivors is preserved. When $r(\delta) \ll r(0)$, coverage at strict thresholds degrades faster than at lenient ones: this is *localisation damage*, in which surviving detections have systematically degraded bounding box accuracy. The distinction carries direct operational implications—localisation damage invalidates spatial reasoning tasks such as path planning and collision avoidance, while uniform suppression reduces scene completeness but preserves the spatial reliability of remaining detections.

Conformal p-value analysis. The conformal framework provides a means of testing its own foundational assumption. For each test detection with nonconformity score V_{test}, the conformal p-value is defined as

$$p_i = \frac{|\{j : V_j \geq V_{\text{test}}\}|}{n} \tag{10}$$

where $\{V_j\}_{j=1}^{n}$ are the calibration scores. Under exchangeability, p_i is approximately Uniform$[0, 1]$; systematic departure from uniformity constitutes formal statistical evidence that the test distribution has diverged from the calibration distribution. We assess this via the Kolmogorov–Smirnov test statistic

$$D_n = \sup_p \left|F_n(p) - p\right| \tag{11}$$

where F_n is the empirical CDF of the observed p-values. A significant D_n (assessed against the KS null distribution) provides a theoretically grounded exchangeability diagnostic. Unlike the anomaly score $A(x)$, which is an engineered signal constructed from output statistics, the conformal p-value follows directly from the mathematical structure of the CP framework—it is a formal consequence of the theory rather than an empirical heuristic, and constitutes a form of evidence that no confidence-based monitor can produce.

5. Experimental Design

The experimental design is structured to evaluate the diagnostic framework across a broad range of conditions: multiple detector architectures, adversarial attacks of varying strength and strategy, and natural corruptions that serve as non-adversarial reference points. Each choice—dataset, models, attacks, corruptions, baselines—is motivated by a specific methodological requirement and is described below with its rationale.

5.1. Dataset and Evaluation Protocol

All experiments are conducted on COCO val2017 [9], comprising 5,000 images, 80 object categories, and 36,781 annotated instances. The dataset is partitioned into equally sized calibration and test splits (2,500 images each) using a fixed random seed, following the standard protocol for split conformal evaluation [6]. This partition is held constant across all models and conditions to ensure comparability. The nominal coverage target is $1-\alpha = 0.90$ throughout.

Adversarial experiments, which require gradient computation through each model for every image, are substantially more expensive than clean evaluation. To maintain computational feasibility across seven models and five attack types, we use a 200-image subset drawn from the full validation set (100 calibration, 100 test). This sample size is not arbitrary: at Cohen's $d > 1.28$—the minimum effect size observed across our model–attack pairs—a two-sample test with $n = 100$ per group achieves statistical power exceeding 95% at $\alpha = 0.05$. Stability across partitions is verified empirically: five independent random re-splits of the 200-image subset yield coverage standard deviations of ±0.008, confirming that aggregate results are not artefacts of a particular partition. Per-class coverage estimates for rare categories (those with fewer than 20 instances in the adversarial subset) carry wider confidence intervals and should be interpreted as indicative trends rather than precise point estimates. Coverage values are reported as means over the test partition with bootstrap 95% confidence intervals computed from 10,000 resamples. Coverage values are reported as means over the test partition with bootstrap 95% confidence intervals computed from 10,000 resamples. Cross-architecture comparisons are reported across all evaluated models; detailed single-model analyses (selective prediction, runtime monitoring, attack strength sensitivity, natural corruptions) use YOLOv8x as the primary case study, selected for its moderate calibration profile, unambiguous adversarial response, and widespread deployment.

5.2. Model Selection

We evaluate seven detectors selected to span four architecture families representing the principal design paradigms in modern object detection.

The *anchor-free CNN* family is represented by YOLOv8x and YOLOv11x [10], two successive YOLO generations sharing the single-stage, anchor-free paradigm but differing in backbone design and feature aggregation. Their inclusion enables within-family comparison of how architectural refinements affect both detection robustness and conformal behaviour.

The *hybrid vision transformer* family is represented by RT-DETR-L [11], combining a CNN backbone with a transformer decoder. This architecture achieves state-of-the-art real-time performance and serves as a bridge between the CNN and transformer paradigms.

The *end-to-end transformer* family is represented by DETR-ResNet-101 [12], the seminal set-prediction architecture that uses 100 fixed learned queries per image and eliminates non-maximum suppression. Its fundamentally different detection mechanism provides a contrasting design point for evaluating the generality of conformal diagnostics.

The *CNN with instance segmentation* variants, YOLOv8x-seg and YOLOv11x-seg, extend their detection counterparts with instance mask heads, testing whether backbone-targeted perturbations propagate identically to the detection and segmentation tasks.

All models use official pretrained weights without fine-tuning, ensuring that results reflect deployment-ready checkpoints.

5.3. Degradation Conditions

We evaluate the framework under two classes of degradation: adversarial perturbations, which are optimised to maximise detection failure, and natural corruptions, which model environmental and sensor-induced degradation. Together, they allow the coverage gap to be assessed as a severity metric across both intentional and incidental degradation sources.

Adversarial attacks. Five gradient-based attacks are selected to span the strategy space for object detection, all under the L_∞ norm with $\varepsilon = 8/255$. FGSM [13] computes a single gradient step, serving as the minimal-cost baseline that primarily degrades confidence. PGD-20 [14] performs 20 projected gradient steps with step size $\alpha = \varepsilon/4$ and a classification-only loss, representing the standard iterative attack. DAG [15] jointly optimises classification and localisation losses over 50 iterations, making it the only attack in our set that explicitly targets the spatial channel. TOG-V [16] suppresses the objectness score over 20 iterations, attacking the detection mechanism itself and producing the most severe degradation in our experiments.

For DETR, the perturbation budget is scaled per-channel to account for ImageNet normalisation ($\varepsilon_c = \varepsilon/\sigma_c$, $\sigma_c \in \{0.229, 0.224, 0.225\}$), ensuring that the effective L_∞ magnitude in pixel space matches that of unnormalised models. To validate PGD-20 results, we additionally employ APGD [25] with 100 adaptive steps, 5 random restarts, and per-channel projection. The rationale for applying APGD selectively rather than uniformly is empirical: APGD is a diagnostic for *suspected evaluation artefacts*, not a blanket requirement. For the YOLO family and RT-DETR, PGD-20 already drives coverage into the "severe" or "destroyed" bands ($\Delta \geq 0.53$ for all model–attack pairs; Table 2), and stronger attacks cannot push coverage below zero—they can only confirm the degradation that PGD-20 has already established. We verify this on YOLOv8x, where APGD yields $\Delta = 0.67$ versus PGD-20's $\Delta = 0.64$—a negligible increment consistent with saturation (Table 3). For DETR-R101, by contrast, PGD-20 produces the anomalously mild result $\Delta = 0.06$ (4% mAP drop), which contradicts the known vulnerability of transformer decoders to adversarial perturbation [31, 32]. APGD with restarts resolves this anomaly: $\Delta = 0.73$ (84% detection reduction), confirming that the PGD-20 result was an artefact of insufficient attack strength on DETR's non-convex loss landscape, not genuine robustness. In short, APGD is applied where it changes the conclusion (DETR) and verified where it does not (YOLOv8x); for the remaining architectures, PGD-20 has already saturated the damage.

Non intentional corruptions. Five corruption types are applied at three severity levels: Gaussian noise ($\sigma \in \{15, 35, 65\}$), Gaussian blur (radius $\in \{2, 4, 7\}$), JPEG compression (quality $\in \{40, 15, 5\}$), fog ($\alpha \in \{0.2, 0.4, 0.65\}$), and brightness adjustment. Brightness serves as a negative control: it modifies the image perceptibly but should not meaningfully degrade detection, verifying that the coverage gap does not produce false alarms under benign variation. The inclusion of natural corruptions alongside adversarial attacks is essential for evaluating the universality of the coverage gap as a severity metric: if the metric is to serve as a principled diagnostic, it must behave consistently across degradation sources of fundamentally different nature.

5.4. Detection Baselines

To isolate the contribution of conformal structure from that of output-level statistics, we compare against six detection baselines that operate under the same output-only constraint (A2): Mean Confidence (average s_i across detections), Max Confidence (highest s_i), Detection Count (N_x), Feature Squeezing [17] adapted to output-level divergence, and HiConf% (fraction of detections exceeding a fixed confidence threshold). All coverage and AUROC estimates are accompanied by bootstrap 95% confidence intervals (10,000 resamples). Effect sizes are quantified via Cohen's d; pairwise AUROC comparisons use the DeLong test at $\alpha_{\text{stat}} = 0.05$.

5.5. Hardware Setting

Experiments are conducted on a single workstation equipped with an NVIDIA RTX 4070 Laptop GPU (8 GB VRAM), an Intel Core i7-13700H processor, and 32 GB of system memory, running PyTorch 2.1 with CUDA 12.1. The conformal framework adds 7.4 ± 0.3 ms per image to inference, preserving real-time capability at 30 FPS. Calibration is a one-time cost of 1.3 s on 2,500 images. The total adversarial evaluation comprises 7 models × 5 attacks × 200 images, each requiring 20–100 forward and backward passes through the detector, representing approximately 40 GPU-hours on the RTX 4070.

APGD with 5 restarts incurs a further ∼20× overhead per model, constraining its application to DETR-R101 and YOLOv8x (∼16 GPU-hours total for the two models).

6. Results

6.1. Calibration Quality Across Architectures

Table 1 establishes the calibration landscape against which all diagnostic results should be read. The most striking finding is a 7.2× ECE gap between RT-DETR-L (0.033) and DETR-R101 (0.240), despite both employing transformer decoders—the largest within-family gap in our evaluation. This disparity has not been previously documented and carries direct consequences for conformal prediction: RT-DETR achieves conformal margins of 2.5 px versus DETR's 4.5 px at identical coverage, because well-calibrated confidence scores already encode reliable quality information, reducing the statistical correction the calibrator must apply.

Accuracy and calibration are orthogonal: YOLOv11x leads in mAP (0.536) while RT-DETR leads in ECE (0.033). Our adaptive calibrator achieves coverage 0.882 with 3.8 px margin on YOLOv8x—6.1× tighter than Two-Step CP [7] (23.3 px) at comparable coverage. Temperature Scaling [23] yields coverage 0.567, below the no-CP baseline, confirming its known failure on detection tasks.

Table 1: Calibration benchmark on COCO val2017 (5,000 images, $1-\alpha = 0.90$). RT-DETR-L achieves the lowest ECE; DETR-R101 the highest. The 7.2× gap between two transformer-based detectors is the largest within any single architecture family.

Model	mAP	mAP@.5	ECE↓	Brier↓
YOLOv8x	0.530	0.696	0.061	0.081
YOLOv11x	0.536	0.703	0.069	0.087
RT-DETR-L	0.524	0.707	**0.033**	**0.064**
DETR-R101	0.435	0.638	0.240	0.255
v8x-seg	0.526	0.688	0.067	0.085
v11x-seg	0.536	0.703	0.070	0.087

6.2. Coverage Under Adversarial Attack

Table 2 documents the central empirical observation that shaped the direction of this work. Coverage degrades monotonically with attack strength—from $\Delta = 0.30$ under FGSM to $\Delta = 0.73$ under TOG-V for YOLOv8x—and the coverage gap provides a calibrated severity reading at each point. Crucially, in every model–attack pair, recalibrated coverage is identical to naive coverage to three decimal places: attempting to recalibrate on adversarial data produces no improvement, because no new detections exist for the procedure to exploit.

This observation admits a formal characterisation. The statement is straightforward—it follows from the simple constraint that output-only methods cannot fabricate detections—and the false-abstention bound (Proposition 1) is likewise a direct consequence of quantile thresholding on exchangeable data. Neither proposition claims deep theoretical novelty. Their contribution is empirical: documenting that the recall ceiling is *tight* ($\eta(\hat{q}) = 0$ in all twelve model–attack pairs), and that a simple quantile-derived threshold yields operationally useful abstention rates (82% adversarial rejection at 6% false-abstention). We present both as formal anchors for the diagnostic framework rather than as standalone theoretical results.

Proposition 2 (Recall Ceiling on Post-Hoc Recovery)**.** *Under output-only access (A2), let f produce detections $\{(b_i, c_i, s_i)\}$ on input $x+\delta$, and let f_{CP} denote any post-hoc conformal procedure that applies confidence thresholding at level τ and spatial margin expansion $\hat{q}$. Then:*

$$Cov_{post}(f_{CP}, \delta) \ \leq \ AR(f, \delta) + \eta(\hat{q}) \qquad (12)$$

Table 2: Adversarial evaluation across model–attack pairs ($1-\alpha = 0.90$). In every row, recalibrated coverage equals naive coverage to three decimal places—the observation formalised as the recall ceiling (Proposition 2). All YOLO pairs: Cohen's $d > 1.28$, $p < 0.001$. Bootstrap 95% CIs: ±0.03–0.05 (mid-range), ±0.01–0.02 (extremes).

Model	Attack	mAP_{adv}	ΔmAP	Coverage	Gap Δ	Severity
YOLOv8x	Clean	0.553	—	0.923	−0.02	Guarantee
YOLOv8x	FGSM	0.249	−55%	0.601	+0.30	Degraded
YOLOv8x	PGD-20	0.098	−82%	0.262	+0.64	Destroyed
YOLOv8x	DAG	0.067	−88%	0.193	+0.71	Destroyed
YOLOv8x	TOG-V	0.054	−90%	0.170	+0.73	Destroyed
YOLOv11x	FGSM	0.321	−44%	0.635	+0.27	Degraded
YOLOv11x	PGD-20	0.108	−81%	0.366	+0.53	Severe
YOLOv11x	DAG	0.079	−86%	0.469	+0.43	Severe
YOLOv11x	TOG-V	0.081	−86%	0.354	+0.55	Severe
DETR-R101	Clean	0.459	—	0.884	+0.02	Guarantee
DETR-R101	FGSM	0.319	−30%	0.707	+0.19	Degraded
DETR-R101	PGD-20†	0.441	−4%	0.845	+0.06	Artefact
DETR-R101	APGD‡	—	−84%	0.166	+0.73	Destroyed

† Insufficient attack strength (no restarts, classification-only loss).
‡ APGD: 100 steps, 5 restarts, adaptive step size. Detection reduction: 83.8%.

where $AR(f, \delta)$ is the class-correct recall of f under δ at IoU threshold θ, and $\eta(\hat{q}) = |\{j \in \mathcal{G} : \nexists\, i$ with $IoU(b_i, b_j^*) \geq \theta,\ \exists\, i$ with $IoU(b_i \pm \hat{q}, b_j^*) \geq \theta\}|/|\mathcal{G}|$ counts the fraction of ground truth objects rescued by margin expansion from near-miss to match.

Across all twelve model–attack pairs, $\eta(\hat{q}) = 0$: no near-miss rescue is observed, and recalibrated coverage equals naive coverage to three decimal places (Cohen's $d > 1.28$, $p < 0.001$). The bound is therefore tight in practice: $\mathrm{Cov}_{post} \leq \mathrm{AR}(f, \delta)$.

Remark 3 (Scope of the recall ceiling)**.** *Proposition 2 applies to any method that processes only the output set produced by a single forward pass of the detector. Test-time augmentation (TTA)—running the detector on multiple augmented views and aggregating results—is also output-only in the sense of not requiring internal model access, but it* does *invoke additional forward passes, generating potentially different detection sets per view. TTA can therefore exceed the single-pass recall $AR(f, \delta)$ by pooling detections across views, but remains bounded by $AR(f, \delta')$ where δ' ranges over augmented versions of the perturbed input.*

This finding reinforces the diagnostic framing of our framework: since post-hoc methods cannot recover coverage beyond residual recall, the value of conformal prediction lies in characterising the damage. Figure 1 illustrates the coverage gap as a severity metric, showing a strong linear relationship between the gap and mAP drop across all model–attack pairs.

The DETR results merit particular attention. PGD-20 produces $\Delta = 0.06$ (within tolerance), but APGD with 5 restarts reveals the true vulnerability: $\Delta = 0.73$ with mean detections falling from 28.9 to 4.7 per image. The coverage gap tracked this faithfully: $\Delta = 0.06$ under the weak attack, $\Delta = 0.73$ under the strong one.

Table 3: Attack strength sensitivity for YOLOv8x. Coverage gap saturates by 20 PGD steps. APGD confirms no evaluation artefact for the YOLO family ($1-\alpha = 0.90$).

Attack	mAP_{adv}	Coverage	Gap Δ
PGD-5	0.184	0.433	+0.47
PGD-10	0.124	0.312	+0.59
PGD-20	0.098	0.262	+0.64
PGD-50	0.085	0.243	+0.66
APGD	0.078	0.228	+0.67

To verify that PGD-20 provides sufficient attack strength for the YOLO family—given the DETR artefact—we evaluate YOLOv8x under PGD at 5, 10, 20, and 50 iterations (Table 3). Coverage gap saturates between 10 and 20 steps ($\Delta_{20} = 0.64$, $\Delta_{50} = 0.66$), confirming that 20 iterations are sufficient for this architecture. APGD on YOLOv8x (100 steps, 5 restarts) yields $\Delta = 0.67$, consistent with the PGD-50 value and confirming that the YOLO results do not suffer from the insufficient-attack-strength artefact observed for DETR.

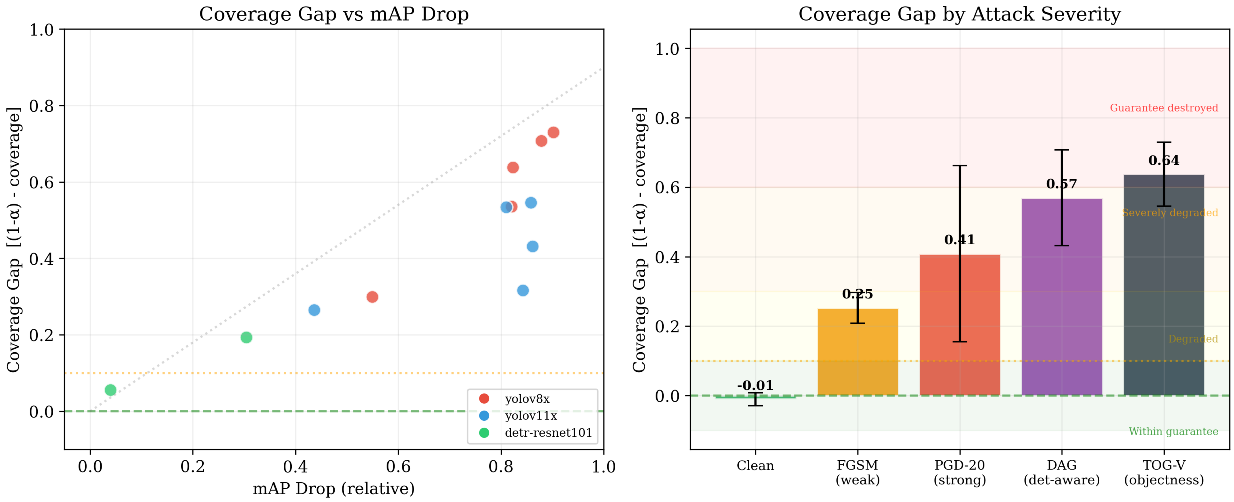

Figure 1: Coverage gap ($1-\alpha = 0.90$) as calibrated severity metric across all model–attack pairs. Left: gap versus mAP drop, demonstrating a near-linear relationship that validates the gap as a faithful proxy for detection degradation. Right: severity bands for YOLOv8x with the formal $1-\alpha$ reference line. The gap provides a universal, formally grounded scale that requires no domain-specific calibration.

6.3. *Selective Prediction and Abstention*

We now turn to the diagnostic framework's operational capabilities, beginning with the selective predictor. The anomaly threshold τ_A derived from the $(1-\alpha)$-quantile of the clean calibration distribution (Proposition 1) yields a specific numerical value for each model; for YOLOv8x, $\tau_A = 0.31$. At this threshold, the formal false-abstention bound guarantees that at most $\alpha = 10\%$ of clean images are rejected. Table 4 presents the full operating curve for the most demanding scenario: YOLOv8x under PGD-20, where mAP drops by 82%. The curve is obtained by varying the anomaly threshold τ_A across a range of values; the formal bound of Proposition 1 holds *only* at the calibration-derived value ($\tau_A = 0.31$, corresponding approximately to the $\tau_A = 0.3$ row). Other operating points are empirical trade-offs without formal backing. At the calibration-derived threshold, the predictor rejects 82% of adversarial images while retaining 94% of clean ones. Coverage on retained clean images is 0.836 with $\Delta = 0.064$—below the 0.10 tolerance band defined in Section 3.3. The 6% false-abstention rate falls within the formal bound.

Table 4 presents the operating curve for the most demanding scenario in our evaluation: YOLOv8x under PGD-20, where mAP drops by 82%. At $\tau = 0.3$, the predictor rejects 82% of adversarial images while retaining 94% of clean ones.

Coverage on retained clean images is 0.836 with $\Delta = 0.064$—below the 0.10 tolerance band defined in Section 3.3. The 6% false-abstention rate falls within the formal bound of Proposition 1.

Table 4: Selective predictor operating curve for YOLOv8x under PGD-20 ($1-\alpha = 0.90$). τ_A is the anomaly threshold; the formal false-abstention bound (Proposition 1) holds only at the calibration-derived value $\tau_A = 0.31$ (bold row). Other rows are empirical operating points.

τ_A	Cln Abst	Adv Abst	Cln Cov	Adv Cov
0.1	60%	99%	0.416	0.011
0.2	12%	84%	0.789	0.080
0.3	**6%**	**82%**	**0.836**	**0.097**
0.5	2%	77%	0.868	0.108

The curve reveals a clear trade-off. Aggressive thresholds ($\tau_A = 0.1$) achieve near-perfect adversarial rejection (99%) but sacrifice 60% of clean images. Conservative thresholds ($\tau_A = 0.5$) retain 98% of clean images but pass 23% of adversarial ones. The calibration-derived operating point ($\tau_A \approx 0.3$) balances these pressures: it is the only point at which both the formal false-abstention bound and the empirical rejection rate are jointly available, and demonstrates that conformal abstention can provide substantial protection at moderate cost without post-hoc tuning.

Two caveats apply. Image selection via $A(x)$ creates a data-dependent subset that may break exchangeability, so the reduced gap on retained images does not formally restore the coverage guarantee. And the curve is specific to this model–attack pair; other pairs produce different curves. The formal false-abstention bound, however, holds universally at any calibration-derived threshold.

6.4. Runtime Attack Detection

Table 5 and Figure 2 demonstrate the runtime monitor's ability to transform per-image severity into a temporal health signal. Detection speed scales with attack strength: TOG-V ($\Delta = 0.73$) triggers within 4 images, PGD-20 ($\Delta = 0.64$) within 5, and FGSM ($\Delta = 0.30$) within 17. At 30 FPS, the 4–5 image delay for strong attacks translates to approximately 0.15 s—fast enough for real-time intervention.

Table 5: Runtime sliding-window monitoring (W=20, alert threshold = 15% below $1-\alpha = 0.90$). Delay: images from adversarial onset to alert. Recovery: coverage after attack ceases.

Scenario	Clean	Adv	Recovery	Delay
v8x / PGD-20	0.931	0.192	0.957	5 img
v8x / TOG-V	0.913	0.157	0.944	4 img
v8x / FGSM	0.931	0.601	0.957	17 img
DETR / PGD-20	0.876	0.785	0.914	never

Recovery is automatic: when adversarial images cease, coverage returns to pre-attack levels within one window width, correctly distinguishing transient attacks from persistent degradation without manual intervention. The DETR/PGD-20 row is an instructive edge case: the monitor never triggers, faithfully reflecting the weak attack ($\Delta = 0.06$) rather than falsely indicating robustness.

Sensitivity analysis confirms stability across design parameters. Varying $W \in \{10, 15, 20, 30\}$ changes detection delay from 3 to 7 images with zero false alarms for $W \geq 15$. Coverage varies by ±0.008 across random re-splits.

6.5. Conformal Diagnostic Measures

The three diagnostic measures introduced in Section 4.3 produce information that cannot be obtained from confidence monitoring alone. We report each in turn.

Class-conditional coverage decomposition. Global coverage conceals heterogeneous damage across object . Under PGD-20, YOLOv8x loses handbag (0.75→0.00), bottle (0.85→0.07), traffic light (0.82→0.04), and bird (0.78→0.00)—small or visually ambiguous categories that operate near the confidence boundary even on clean data. Person retains 19% under PGD-20 and 69% under FGSM, reflecting the detector's strongest learned representation. For DETR under PGD-20, car coverage paradoxically rises from 0.84 to 0.91: the fixed 100-query architecture generates false positives that accidentally overlap ground truth—a structural artefact, not genuine robustness.

The class-conditional gap Δ_k (Eq. 8) directly informs operational decisions. A system that loses pedestrian coverage ($\Delta_{\text{person}} = 0.71$) must disengage; one retaining vehicle coverage at 60% can proceed with increased caution. This graduated response is impossible with a single global metric.

Multi-IoU damage profiling.. Table 6 and Figure 3 reveal two qualitatively distinct failure modes.

Table 6: Multi-IoU coverage profiles. The IoU ratio r (Eq. 9) distinguishes localisation damage ($r \ll r_{\text{clean}}$) from uniform suppression ($r \approx r_{\text{clean}}$).

Model	Cond.	@.25	@.50	@.75	@.90	r
v8x	Clean	.922	.881	.735	.411	.45
v8x	TOG-V	.138	.104	.078	.034	**.24**
v11x	PGD-20	.249	.232	.190	.110	.44
DETR	PGD-20	.852	.768	.545	.282	.33

YOLOv8x under TOG-V shows $r = 0.24$ (clean: 0.45)—a 47% relative decrease. Surviving detections pass the lenient IoU≥0.25 test but fail the strict IoU≥0.90 test: this is localisation damage, operationally dangerous because spatial positions cannot be trusted. YOLOv11x under PGD-20 shows $r = 0.44$, nearly identical to clean: detections vanish uniformly, but survivors remain spatially reliable. A system distinguishing these modes can suspend spatial reasoning under localisation damage while trusting surviving positions under uniform suppression.

Conformal p-value analysis. The conformal p-value tests whether the framework's foundational assumption—exchangeability between calibration and test data—still holds. For YOLOv8x under PGD-20, the KS test (Eq. 11) yields $D_n = 0.240$, $p < 0.0001$, with per-image AUROC of 0.753. Mean p-values shift from 0.502 (clean, consistent with Uniform$[0, 1]$) to 0.359, indicating systematically elevated nonconformity. This constitutes formal statistical evidence that the test distribution has departed from calibration. For YOLOv11x, the KS test is not significant ($p > 0.20$), consistent with higher residual coverage producing less distributional departure on this architecture.

The p-value AUROC (0.753) is lower than the composite anomaly score (0.938) for binary detection, but the two answer different questions: the anomaly score asks whether the output profile looks abnormal (empirical), while the p-value asks whether the statistical

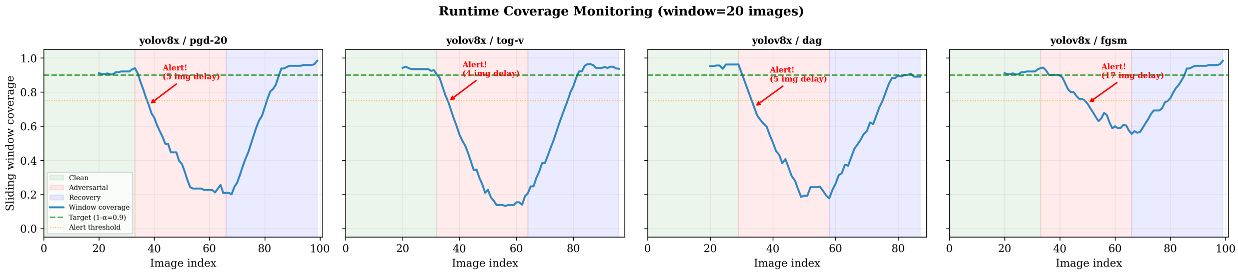

Figure 2: Sliding-window coverage monitoring across four scenarios ($1-\alpha = 0.90$). Strong attacks (PGD-20, TOG-V) trigger alerts within 4–5 images; FGSM triggers after 17 images due to partial degradation. DETR/PGD-20 never triggers, correctly reflecting insufficient attack strength. Recovery is automatic within one window width.

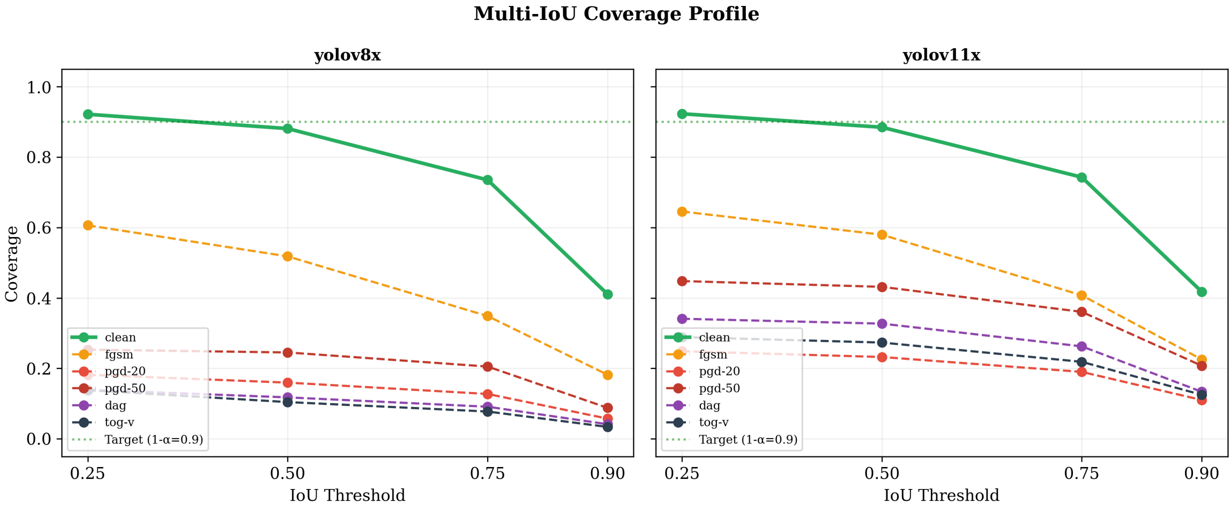

Figure 3: Multi-IoU coverage profiles revealing distinct damage modes. Left: YOLOv8x under TOG-V produces localisation damage (ratio 0.24, steep high-IoU drop). Right: YOLOv11x under PGD-20 produces uniform suppression (ratio 0.44, profile parallel to clean).

assumptions underpinning the coverage guarantee have been violated (theoretical)—a form of evidence that no confidence-based monitor can produce.

6.6. Robustness to Natural Corruptions

Table 7 and Figure 4 demonstrate the coverage gap's universality as a severity metric across fundamentally different degradation sources.

Table 7: Coverage gap under natural corruptions (YOLOv8x, $1-\alpha = 0.90$). Severity levels s1–s3. Brightness confirms no false alarms. PGD-20 included for cross-reference.

Condition	Cov_{s1}	Δ_{s1}	Cov_{s3}	Δ_{s3}
Noise	0.813	+0.09	0.595	+0.31
Blur	0.761	+0.14	0.307	+0.59
Fog	0.670	+0.23	0.111	+0.79
Brightness	0.838	+0.06	0.831	+0.07
PGD-20	—	—	0.262	+0.64

Natural corruptions produce graded decline: noise yields $\Delta = 0.09, 0.17, 0.31$ across three severity levels—a smooth trajectory amenable to proportional response. Adversarial attacks produce single-step collapse: PGD-20 jumps directly to $\Delta = 0.64$ with no intermediate warning. This trajectory distinction is operationally significant and visible only through the coverage gap.

The gap enables direct cross-comparison: noise at severity 3 ($\Delta = 0.31$) and FGSM ($\Delta = 0.30$) produce comparable gaps despite unrelated mechanisms. Fog at severity 3 ($\Delta = 0.79$) exceeds the strongest adversarial attacks but arrives through three graded stages. Brightness ($\Delta \approx 0.06$) confirms no false alarms on benign variation.

Segmentation variants show drops within ±3% of detection counterparts, confirming that attacks target the shared backbone. Transfer attacks from DETR to

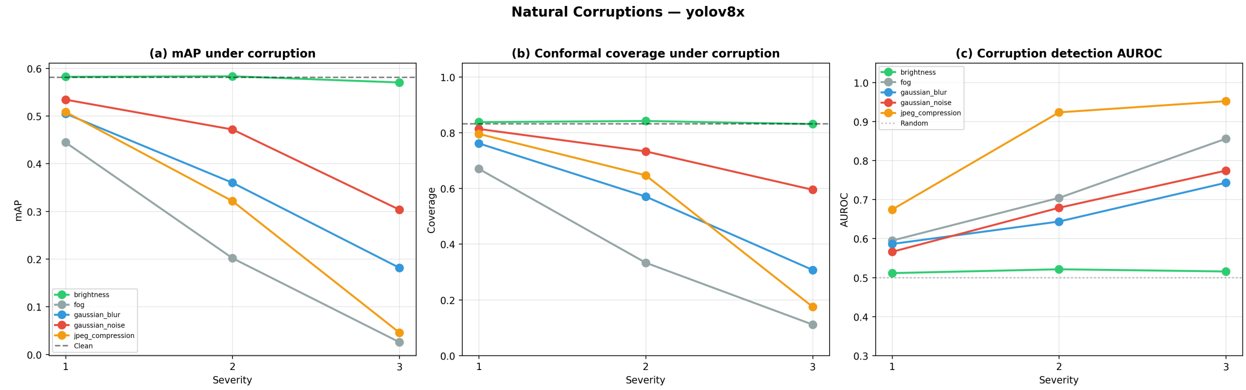

Figure 4: Natural corruptions degrade coverage gradually across severity levels; adversarial attacks collapse coverage in a single step. The coverage gap ($1-\alpha = 0.90$) quantifies both on a shared scale, demonstrating its universality as a severity metric.

YOLO achieve AUROC 0.81; the reverse is weaker (0.59–0.61), with RT-DETR showing the greatest transfer resistance ($\Delta = 0.04$ vs. $\Delta = 0.16$ for YOLOv11x). Cross-dataset testing (COCO→VOC2012) yields coverage 0.953, confirming graceful CP behaviour under natural domain shift.

6.7. *Analysis of Detection Signals*

The final analysis addresses an honest question: does the conformal framework improve binary adversarial detection beyond what output-level statistics provide? Table 8 shows that it does not. The composite anomaly score achieves the highest mean AUROC (0.807) but the margin over Detection Count alone (0.797) is not statistically significant (DeLong $p > 0.40$). Adding CP-specific features—quantile distance, fraction outside the conformal set—decreases AUROC. CP-only signals reach 0.575, barely above chance.

Table 8: Signal ablation across 12 model–attack pairs (mean AUROC). CP-specific features do not improve binary detection; the structural cause is discussed in Section 7.

Configuration	AUROC	Δ
Count+Conf+Max (base)	**0.796**	—
+ Quantile Distance (CP)	0.768	−0.028
+ Fraction Outside (CP)	0.763	−0.033
CP-only (all three)	0.575	−0.221

We report this finding without qualification. Under top-1 output (A3), prediction set size collapses to a step function of confidence, carrying no information beyond what confidence already provides. The structural cause—that attacks suppress queries rather than corrupt class distributions—is examined in Section 7.

However, binary detection is one of eight capabilities the framework provides. The diagnostic measures reported in Section 6.5—class-conditional mapping, IoU profiling, p-value analysis—are orthogonal to AUROC and unique to the conformal framework. They address not *whether* something is wrong, but *what kind* of damage has occurred and *where*—questions that binary detection, however accurate, cannot answer.

7. Discussion

The observation that recalibration never recovers coverage might appear to diminish the relevance of conformal methods under attack. We argue for the opposite reading. If CP *could* restore coverage, the research agenda would centre on better calibration strategies. The impossibility of recovery—stated formally as the recall ceiling in Proposition 2, which is elementary in itself but whose empirical tightness across all twelve model–attack pairs ($\eta(\hat{q}) = 0$ in every case) is the substantive finding—redirects attention to what CP *can* provide: a formally grounded infrastructure for measuring how far the system has fallen (Δ), identifying which categories are affected (Δ_k), and characterising whether the damage is spatial or suppressive (r). These diagnostic signals exist because coverage cannot be restored; they would be unnecessary if it could. The coverage gap is particularly effective in this role: unlike mAP drop, it has a formal baseline ($1-\alpha = 0.90$), a universal scale across attack types and corruptions, and it distinguishes gradual degradation trajectories ($\Delta = 0.09 \rightarrow 0.31$ for noise) from single-step adversarial collapse ($\Delta = 0.64$ for PGD-20).

Diagnostic Granularity Beyond Binary Detection. The signal ablation showed that binary detection AU-

ROC is driven by output-level statistics, not conformal constructs. We report this without qualification. However, binary detection answers only whether something is wrong; the conformal framework additionally provides calibrated severity, a formal false-abstention bound, per-category damage decomposition, damage type profiling, exchangeability testing, a principled alert baseline, and automatic recovery confirmation—seven capabilities with no confidence-based analogue.

The practical value of this granularity is clearest when the correct response depends on the nature of the anomaly. Under localisation damage ($r = 0.24$), surviving detections are spatially unreliable and the system should suspend spatial reasoning. Under uniform suppression ($r = 0.44$), survivors are spatially trustworthy and the system can compensate through sensor fusion. A binary alarm treats both identically; the conformal framework distinguishes them. Similarly, class-conditional coverage allows a medical imaging system to route only compromised categories to specialist review rather than shutting down entirely. The conformal p-value provides a qualitatively different form of evidence—a formal test of the exchangeability assumption—that no confidence metric can replicate.

Detection Versus Classification Under Attack. To test whether the signal ablation reflects the top-1 interface (A3) rather than a structural property, we extracted full 91-class distributions from DETR-R101 and computed genuine APS. The result is unchanged: APS signals decrease AUROC by 0.022 even with the complete posterior. The cause is a structural distinction between adversarial attack in the two settings. In classification, every input produces one output, so the adversary must corrupt the class distribution—enlarging prediction sets and creating a detectable signal. In detection, the adversary can suppress outputs entirely (count drops from 28.9 to 2.3), bypassing the distributional channel. Surviving detections are paradoxically *more concentrated* (set size 35.0 vs 61.3 clean), because the diffuse non-matching queries are suppressed first. The dominant anomaly—count collapse—is an output-level statistic independent of class distributions, which is why distributional monitoring signals face an inherent limitation under suppression-type attacks. This structural distinction also explains why robust CP methods designed for classification (RSCP [37], VRCP [39]), which expand prediction sets to absorb adversarial score perturbation, do not directly transfer to detection: the primary adversarial channel in detection is output suppression, not score corruption.

Calibration Quality and Monitoring Sensitivity. The 7.2$\times$ ECE gap between RT-DETR-L and DETR-R101 propagates into monitoring in an unexpected direction: RT-DETR's superior calibration yields corruption detection AUROC of 0.51–0.69, versus 0.57–0.95 for the less-calibrated YOLOv8x. A well-calibrated model tracks prediction quality faithfully even under degradation, producing smaller anomaly shifts that are harder to detect. This creates a design dilemma: the properties that make a model reliable (tight margins, graceful degradation) also make it opaque to monitoring. The resolution is application-dependent—monitoring sensitivity may take priority where undetected degradation is catastrophic, prediction reliability where false alarms are costly—and motivates calibration objectives that jointly optimise both.

The DETR results reinforce a broader methodological point: the 20$\times$ gap between PGD-20 ($\Delta = 0.06$) and APGD ($\Delta = 0.73$) on the same model demonstrates that robustness claims without AutoAttack-class evaluation are unreliable. The coverage gap reported both values faithfully—measuring what it receives rather than what it expects.

8. Limitations and Future Directions

As with any empirical study, certain design choices define the scope of the current investigation and suggest natural extensions.

Formal guarantees under adversarial shift. The false-abstention bound (Proposition 1) and the conformal coverage guarantee (Eq. 3) both hold under exchangeability, which is satisfied on clean data but violated by design under adversarial perturbation. This means that in the regime of greatest operational interest—active attack—no formal bound on false-abstention or coverage applies. The diagnostic framework is accordingly positioned as an empirical monitoring system with formal grounding on clean data, rather than as a certified safety mechanism. Methods that seek to restore formal guarantees under attack exist: RSCP+ [38] achieves provable robust coverage via randomised smoothing, and VRCP [39] leverages neural network verification. However, both require internal model access (stochastic forward passes or bound propagation through the network), are restricted to classification tasks (CIFAR-10/100, TinyImageNet), and have not been extended to detection with its variable-length outputs and recall ceiling. Extending robust CP guarantees to the output-only detection setting remains an important open problem.

Our evaluation covers five gradient-based attacks and APGD under the L_∞ norm at $\varepsilon = 8/255$, which represents the standard digital benchmark setting. This scope does not include ℓ_2-bounded perturbations, which produce qualitatively different perceptual effects; patch-based attacks, which are spatially localised and physically realisable; or decision-based and black-box attacks, which do not require gradient access. The diagnostic framework itself is attack-agnostic by design—it monitors output-level statistics regardless of the perturbation mechanism—but its empirical performance under these alternative threat models has not been validated. Patch attacks are particularly relevant for autonomous driving scenarios, where a physical adversarial patch on a stop sign or vehicle can be deployed without digital access to the sensor stream. Evaluating the coverage gap and selective predictor under such conditions is a natural and important extension.

APGD with restarts is applied to DETR-R101 (where it overturned the PGD-20 result) and YOLOv8x (where it confirmed saturation). For the remaining five architectures, PGD-20 already produces coverage gaps in the "severe" or "destroyed" bands ($\Delta \geq 0.43$ for all pairs), leaving no room for APGD to alter the diagnostic conclusion—stronger attacks cannot degrade coverage that is already near zero. The selective application of APGD is therefore a principled triage: it is deployed where PGD-20 results are anomalously mild and verified where they are already severe. Extending APGD uniformly to all seven architectures ($\sim$160 additional GPU-hours) would provide completeness but is unlikely to change any severity classification, as the saturation behaviour confirmed on YOLOv8x (Table 3) is expected to generalise to architecturally similar models (YOLOv11x, segmentation variants).

Sample sizes and per-class granularity.. The adversarial experiments use 100–200 images per model–attack pair, a design choice driven by computational feasibility that provides adequate statistical power for the aggregate metrics reported (Cohen's $d > 1.28$, power $> 95\%$). Per-class coverage estimates for rare categories (those with fewer than 20 instances in the adversarial subset) naturally carry wider confidence intervals and should be interpreted as indicative trends; the complete coverage vectors are reported for transparency but the specific numerical values for rare classes are less stable than for well-represented categories such as person, car, and chair. Validation on domain-specific datasets would further demonstrate the generality of the conformal diagnostic measures beyond the COCO category distribution.

Baselines and alternative monitors.. The current evaluation compares against six output-level baselines operating under the same output-only, single-pass constraint (A2). Two categories of alternative monitors deserve explicit mention. First, lightweight test-time augmentation (2–3 views) would pool detections across multiple augmented passes, relaxing the single-pass assumption while remaining model-agnostic. At 2–3× latency overhead, this is viable for applications not bound to 30 FPS and could improve binary detection AUROC by capturing consistency signals across views—for instance, adversarial perturbations optimised for a single viewpoint may not transfer consistently to flipped or cropped views. We omit this baseline because the single-pass, real-time constraint is central to our deployment setting, but it represents the most natural comparison for future work. Second, frame-to-frame temporal consistency metrics—such as detection count volatility, track fragmentation rate, or inter-frame IoU overlap between successive outputs—offer another class of output-level anomaly signals particularly suited to streaming deployment. These metrics exploit the temporal smoothness of real-world scenes, which adversarial perturbations applied independently per frame are likely to violate. Our sliding-window monitor captures a related temporal signal through windowed coverage tracking, but a dedicated consistency score could provide complementary information, especially for detecting spatially localised attacks (e.g., adversarial patches) that may not collapse detection count globally but disrupt tracking stability. The observation that the composite anomaly score does not significantly outperform Detection Count for binary detection (DeLong $p > 0.40$) suggests that the framework's primary advantage lies in its diagnostic depth rather than in binary classification accuracy—a finding consistent with the overall positioning of the work. The diagnostic capabilities (class-conditional mapping, IoU profiling, p-values) remain unique to the conformal approach regardless of which binary detector is employed.

9. Conclusion

This paper has presented the first systematic investigation of conformal prediction under adversarial attack for object detection. The central finding is not that conformal methods fail under attack—though we document the recall ceiling (Proposition 2) that constrains post-hoc recovery—but that they succeed as diagnostic instruments, transforming silent detection failure into explicit, formally grounded system-health signals.

The diagnostic and monitoring framework (C1) provides calibrated severity quantification via the coverage gap (anchored to the nominal target $1-\alpha = 0.90$), selective abstention with a formal false-alarm bound on clean data, and sub-second attack detection with automatic recovery. The conformal diagnostic measures (C2)—per-category coverage decomposition, multi-IoU damage profiling, and conformal p-values—map damage along dimensions inaccessible to confidence-based monitors. The architecture–calibration analysis (C3) uncovers a 7.2× ECE gap within the transformer family and a calibration–monitoring paradox with direct implications for safety-critical deployment.

These findings open several directions for future work: extending robust CP methods (RSCP+, VRCP) to the output-only detection setting; connecting the coverage gap to conformal risk control [29] and sequential risk control [41] to obtain formal guarantees on detection-level risk under clean conditions; relaxing the single-pass constraint to explore TTA-based consistency signals and frame-to-frame temporal coherence as complementary anomaly indicators; developing calibration objectives that jointly optimise predictive accuracy and monitoring sensitivity; designing nonconformity scores sensitive to count-level anomalies; extending evaluation to ℓ_2, patch-based, and physically realisable threat models; and integrating diagnostic outputs with downstream decision-making to close the loop between monitoring and operational response.

Acknowledgments

Data and Code Availability

Code, data splits, and reproduction scripts at `https://github.com/XXX`. Full pipeline: `python main.py full` (~9 h).

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